4. SYSTEM ANALYSIS AND CLASSIFICATION

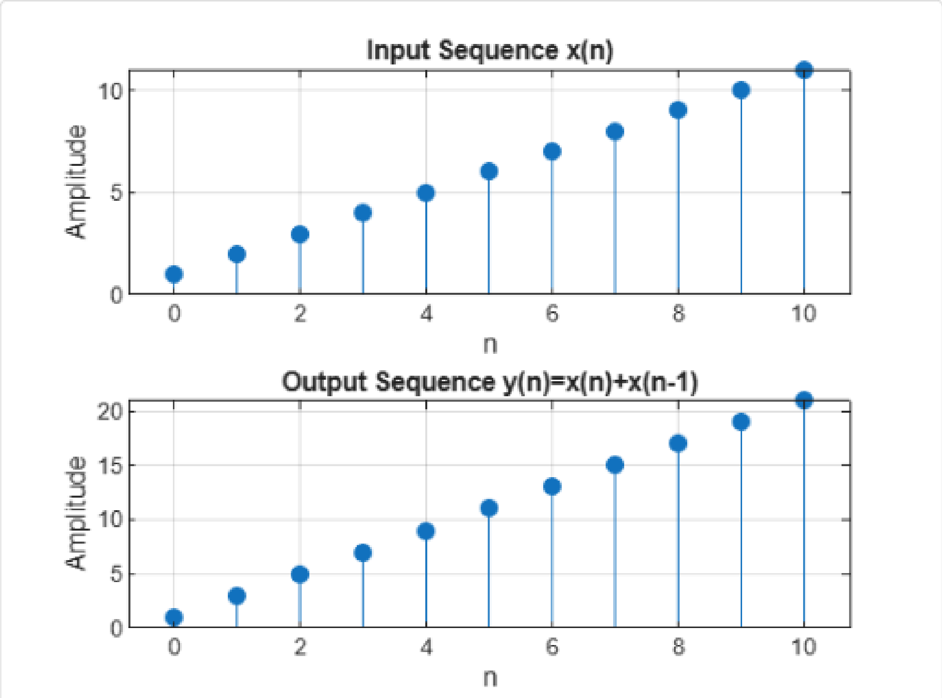
Experiment 4.1

```
% Discrete-Time System
clc;
clear;
close all;
% Input sequence
n = 0:10;

x = [1 2 3 4 5 6 7 8 9 10 11];
% Discrete-Time System
y = filter([1 1],1,x);
% Display output
disp('Output Sequence:');
disp(y);
% Plotting
figure;
subplot(2,1,1);
stem(n,x,'filled');
grid on;
title('Input Sequence x(n)');
xlabel('n');
ylabel('Amplitude');
subplot(2,1,2);
```

```
stem(n,y,'filled');
grid on;
title('Output Sequence y(n)=x(n)+x(n-1)');
xlabel('n');
ylabel('Amplitude');
```

```
Output Sequence:
     1     3     5     7     9    11    13    15    17    19    21
```



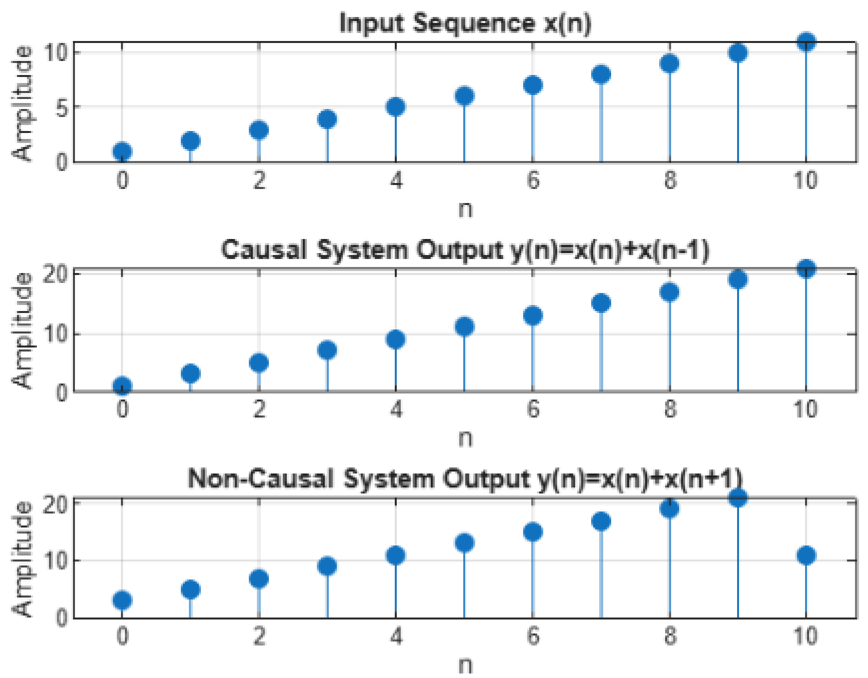
Experiment 4.2

```
% Causal and Non-Causal Systems
clc;
clear;
close all;
% Input sequence
n = 0:10;
x = [1 2 3 4 5 6 7 8 9 10 11];
```

```
% Causal System: y(n)=x(n)+x(n-1)
y_causal = filter([1 1],1,x);
% Non-Causal System: y(n)=x(n)+x(n+1)
y_noncausal = x + [x(2:end) 0];
% Plotting
figure;
subplot(3,1,1);
stem(n,x,'filled');
grid on;
title('Input Sequence x(n)');
xlabel('n');
ylabel('Amplitude');
subplot(3,1,2);
stem(n,y_causal,'filled');
grid on;
title('Causal System Output y(n)=x(n)+x(n-1)');
xlabel('n');
ylabel('Amplitude');
subplot(3,1,3);

stem(n,y_noncausal,'filled');
grid on;
title('Non-Causal System Output y(n)=x(n)+x(n+1)');
xlabel('n');
```

ylabel('Amplitude');



Experiment 4.3

% Analysis of Discrete-Time System

clc;

clear;

close all;

% Input sequence

n = 0:10;

x = [1 2 3 4 5 6 7 8 9 10 11];

% System: y(n) = x(n) + 0.5x(n-1)

b = [1 0.5];

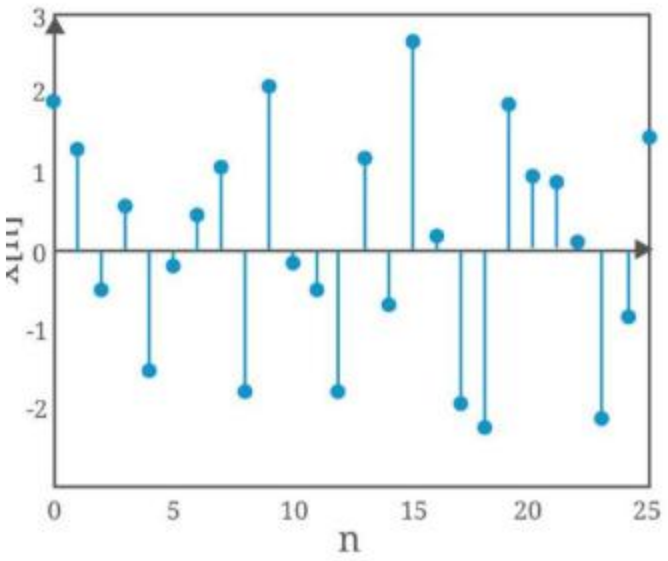
a = [1];

y = filter(b,a,x);

% Display output

```
disp('Input Sequence x(n):');
disp(x);
disp('Output Sequence y(n):');
disp(y);
% Plotting
figure;
subplot(2,1,1);
stem(n,x,'filled');
grid on;
title('Input Sequence x(n)');
xlabel('n');

ylabel('Amplitude');
subplot(2,1,2);
stem(n,y,'filled');
grid on;
title('Output Sequence y(n)');
xlabel('n');
ylabel('Amplitude');
```



Experiment 4.4

```
% Analysis of Continuous-Time System

clc;

clear;

close all;

% Time axis

t = 0:0.01:10;

% Input signal

x = sin(2*pi*t);

% Delayed signal x(t-1)

x_delay = sin(2*pi*(t-1));

% System output

y = x + 0.5*x_delay;

% Plotting

figure;

subplot(2,1,1);
```

```
plot(t,x,'LineWidth',2);
grid on;
title('Input Signal x(t)');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(2,1,2);
plot(t,y,'LineWidth',2);
grid on;
title('Output Signal y(t)=x(t)+0.5x(t-1)');
xlabel('Time (s)');
ylabel('Amplitude');
```